Nuclear Energy

Chapter 4 Unit 2 Guided Notes · Grade 4 Using Natural Resources for Energy

Name: _____ **Date:** _____

BIG QUESTION **What is nuclear power?**

KEY WORDS

nuclear energy — energy released from a material called _____ .

uranium — a _____ material mined from rock.

KEY IDEA

KEY IDEA **Nuclear power uses heat to make electricity — the same pathway as coal.**

Nuclear energy releases a lot of heat. That heat boils water into steam.

ELECTRICITY PATHWAY

NUCLEAR HEAT

↓

BOIL WATER INTO STEAM

↓

SPIN A TURBINE

↓

TURN A GENERATOR

↓

ELECTRICITY

CHECK YOUR THINKING

How is a nuclear power plant like a coal power plant?

Benefits and Limitations

Name: _____ **Date:** _____

BENEFITS AND LIMITATIONS

BENEFIT	LIMITATION
reliable electricity	_____
little air pollution from the plant	_____
nonrenewable (uranium is rare)	_____
dangerous used fuel must be stored safely	_____

SAFETY FIRST

Nuclear power must be controlled carefully. Used fuel can stay dangerous for a very long time and must be stored safely.

The heat must be _____ so the reactor stays safe.

Used fuel is placed in _____ storage.

CHECK YOUR THINKING

What is one benefit AND one limitation of nuclear power?

CONNECT *Chapter 4 connection: nuclear and coal power plants share the same heat → steam → turbine → generator pathway.*